

Finite Difference Methods for Option Pricing

— *Lecture Notes: Three Sessions of 50 Minutes* —

Topics Covered:

Forward / Backward / Central Differences · Explicit FDM · Implicit
FDM

Crank–Nicolson · Leapfrog · von Neumann Stability Analysis

Consistency · Stability · Convergence (Lax–Wendroff)

Connection to Binomial / Trinomial Trees

April 5, 2026

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1 Motivation: Why Finite Difference Methods?

1.1 When Closed-Form Solutions Fail

The Black–Scholes formula gives an analytical price for European options, but in many realistic situations no closed form is available:

- **American options** – early exercise feature
- **Barrier / lookback options** – path-dependent payoffs
- **Options on dividend-paying stocks** (Schwartz 1977)
- **Stochastic-volatility models** (Heston, SABR, ...)

Three broad numerical approaches exist:

1. **Tree models** (Binomial/Trinomial) – discretise the stock-price process.
2. **Monte Carlo simulation** – simulate paths.
3. **Finite Difference Methods (FDM)** – discretise the PDE directly.

Core Idea of FDM

Replace every continuous derivative in the PDE by an algebraic approximation on a finite grid, then solve the resulting system of equations by backward induction.

1.2 The Black–Scholes PDE

For a European option with value $V(S, t)$:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad (1)$$

with terminal condition $V(S, T) = \max(S - K, 0)$ (call) or $\max(K - S, 0)$ (put).

Boundary conditions (call):

$$V(0, t) = 0, \quad V(S, t) \approx S - Ke^{-r(T-t)} \text{ as } S \rightarrow \infty.$$

1.3 Logarithmic Transformation

The coefficients S^2 and S in (1) vary with the state variable, complicating finite-difference stencils. Setting

$$y = \ln S, \quad W(y, t) = V(S, t)$$

and applying the chain rule:

$$\frac{\partial V}{\partial S} = \frac{1}{S} W_y = e^{-y} W_y, \quad (2)$$

$$\frac{\partial^2 V}{\partial S^2} = \frac{\partial}{\partial S} \left(\frac{W_y}{S} \right) = -\frac{W_y}{S^2} + \frac{W_{yy}}{S^2} = e^{-2y} (W_{yy} - W_y), \quad (3)$$

$$\frac{\partial V}{\partial t} = W_t. \quad (4)$$

Substituting into (1) (recall $S = e^y$):

$$\frac{1}{2}\sigma^2 e^{2y} \cdot e^{-2y}(W_{yy} - W_y) + r e^y \cdot e^{-y} W_y + W_t - r W = 0.$$

Log-Transformed BS Equation

$$\frac{1}{2}\sigma^2 W_{yy} + \left(r - \frac{1}{2}\sigma^2\right) W_y + W_t - r W = 0. \quad (5)$$

All coefficients are now *constants*, independent of S .

2 Finite Difference Approximations

2.1 Taylor Expansion Foundation

Let f be sufficiently smooth. Then:

$$f(x_0 + h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(x_0)}{6}h^3 + \dots, \quad (6)$$

$$f(x_0 - h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(x_0)}{6}h^3 + \dots. \quad (7)$$

2.2 Forward Difference (First-Order Accurate)

From (6):

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}, \quad \text{error} = O(h). \quad (8)$$

Uses values at x_0 (known) and $x_0 + h$ (future); truncation error $O(h)$.

2.3 Backward Difference (First-Order Accurate)

From (7):

$$f'(x_0) \approx \frac{f(x_0) - f(x_0 - h)}{h}, \quad \text{error} = O(h). \quad (9)$$

2.4 Central Difference (Second-Order Accurate)

Subtracting (7) from (6):

$$f(x_0 + h) - f(x_0 - h) = 2hf'(x_0) + \frac{h^3}{3}f'''(x_0) + \dots$$

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0 - h)}{2h}, \quad \text{error} = O(h^2). \quad (10)$$

2.5 Second Derivative (Central, Second-Order Accurate)

Adding (6) and (7):

$$f(x_0 + h) + f(x_0 - h) = 2f(x_0) + h^2 f''(x_0) + O(h^4)$$

$$f''(x_0) \approx \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2}, \quad \text{error} = O(h^2). \quad (11)$$

2.6 Grid Setup

For equation (5), define:

- h : spatial step (log-price direction).
- k : time step.
- $W_{i,j} \approx W(ih, jk)$: approximate value at grid node (ih, jk) .

Boundary condition: at maturity $j = J$, $W_{i,J} = \max(e^{ih} - K, 0)$ (call).

Time direction: backward induction from $j = J$ down to $j = 0$.

3 Explicit Finite Difference Method

3.1 Discretisation

In the explicit scheme the spatial derivatives are evaluated at the *known* time level $j + 1$, while only $W_{i,j}$ is unknown:

$$W_y \approx \frac{W_{i+1,j+1} - W_{i-1,j+1}}{2h} \quad (\text{central difference at } j + 1), \quad (12)$$

$$W_{yy} \approx \frac{W_{i+1,j+1} - 2W_{i,j+1} + W_{i-1,j+1}}{h^2} \quad (\text{central difference at } j + 1), \quad (13)$$

$$W_t \approx \frac{W_{i,j+1} - W_{i,j}}{k} \quad (\text{backward in time}). \quad (14)$$

Substituting into (5) and multiplying through by k :

$$\frac{1}{2}\sigma^2 \left(\frac{W_{i+1,j+1} - 2W_{i,j+1} + W_{i-1,j+1}}{h^2} \right) + \left(r - \frac{1}{2}\sigma^2 \right) \left(\frac{W_{i+1,j+1} - W_{i-1,j+1}}{2h} \right) + \frac{W_{i,j+1} - W_{i,j}}{k} - rW_{i,j} = 0.$$

Collecting $W_{i,j}$ on the left:

Explicit Scheme – Brennan and Schwartz 1978

$$(1 + rk) W_{i,j} = a W_{i-1,j+1} + b W_{i,j+1} + c W_{i+1,j+1}, \quad (15)$$

where

$$\begin{aligned} a &= \left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 - \frac{1}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] k, \\ b &= \left[1 - \left(\frac{\sigma}{h} \right)^2 k \right], \\ c &= \left[\frac{1}{2} \left(\frac{\sigma}{h} \right)^2 + \frac{1}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] k. \end{aligned} \quad (16)$$

Verification: $a + b + c = 1$. Direct algebra using (16):

$$a + b + c = \left[\frac{\sigma^2 k}{h^2} - \frac{k}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] + \left[1 - \frac{\sigma^2 k}{h^2} \right] + \left[\frac{\sigma^2 k}{h^2} + \frac{k}{2h} \left(r - \frac{1}{2}\sigma^2 \right) \right] \cdot \frac{1}{2} + \dots$$

Careful grouping yields $a + b + c = 1$. □ □

3.2 Jump-Process Interpretation

Dividing (15) by $1 + rk$:

$$W_{i,j} = \frac{1}{1 + rk} (p^- W_{i-1,j+1} + p W_{i,j+1} + p^+ W_{i+1,j+1}), \quad (17)$$

with $p^- = a$, $p = b$, $p^+ = c$.

This is exactly the *risk-neutral expectation formula* for a three-point jump process (Cox & Ross 1976): the option today is the discounted expected value under the measure (p^-, p, p^+) .

Explicit FDM $\leftrightarrow$ Trinomial Tree

Equation (17) is numerically identical to a *trinomial tree* with jumps $\{-h, 0, +h\}$ and risk-neutral probabilities (p^-, p, p^+) under log-price $y = \ln S$. This is the central synthesis result of Brennan & Schwartz (1978).

3.3 Stability Conditions for the Explicit Scheme

For p^-, p, p^+ to be interpretable as probabilities, all must be non-negative.

Condition on $b \geq 0$:

$$1 - \left(\frac{\sigma}{h}\right)^2 k \geq 0 \implies k \leq \frac{h^2}{\sigma^2}.$$

Condition on $a \geq 0$ (equivalently $c \geq 0$):

$$\frac{1}{2} \left(\frac{\sigma}{h}\right)^2 \geq \frac{1}{2h} \left|r - \frac{1}{2}\sigma^2\right| \implies h \leq \frac{\sigma^2}{\left|r - \frac{1}{2}\sigma^2\right|}.$$

Explicit Stability Conditions – Brennan & Schwartz (1978)

$$h \leq \frac{\sigma^2}{\left|r - \frac{1}{2}\sigma^2\right|}, \quad k \leq \frac{\sigma^2}{\left(r - \frac{1}{2}\sigma^2\right)^2}. \quad (18)$$

Conditionally Stable

The explicit method is **conditionally stable**: if h or k violates (18), numerical errors grow exponentially and the solution diverges.

4 Implicit Finite Difference Method

4.1 Discretisation

In the implicit scheme, the spatial derivatives are evaluated at the *unknown* time level j :

$$W_y \approx \frac{W_{i+1,j} - W_{i-1,j}}{2h}, \quad (19)$$

$$W_{yy} \approx \frac{W_{i+1,j} - 2W_{i,j} + W_{i-1,j}}{h^2}, \quad (20)$$

$$W_t \approx \frac{W_{i,j+1} - W_{i,j}}{k}. \quad (21)$$

Substituting into (5) and rearranging:

Implicit Scheme

$$aW_{i-1,j} + bW_{i,j} + cW_{i+1,j} = (1 + rk)W_{i,j+1}, \quad (22)$$

where

$$\begin{aligned} a &= -\left[\frac{1}{2}\left(\frac{\sigma}{h}\right)^2 + \frac{1}{2h}\left(r - \frac{1}{2}\sigma^2\right)\right]k, \\ b &= 1 + \left(\frac{\sigma}{h}\right)^2 k, \\ c &= -\left[\frac{1}{2}\left(\frac{\sigma}{h}\right)^2 - \frac{1}{2h}\left(r - \frac{1}{2}\sigma^2\right)\right]k. \end{aligned} \quad (23)$$

4.2 Tridiagonal System

For each time step j , equation (22) with $i = 1, \dots, n$ gives the linear system $\mathbf{A}\mathbf{w}_j = \mathbf{f}_j$:

$$\underbrace{\begin{pmatrix} b & c & & \\ a & b & c & \\ & \ddots & \ddots & \ddots \\ & & a & b & c \\ & & & a & b \end{pmatrix}}_{\mathbf{A}} \begin{pmatrix} W_{1,j} \\ \vdots \\ W_{n,j} \end{pmatrix} = (1 + rk) \begin{pmatrix} W_{1,j+1} \\ \vdots \\ W_{n,j+1} \end{pmatrix} - \begin{pmatrix} aW_{0,j} \\ 0 \\ \vdots \\ 0 \\ cW_{n+1,j} \end{pmatrix}.$$

Since $\mathbf{A}$ is independent of j , its LU factorisation is performed **only once**.

4.3 Gaussian Elimination for Tridiagonal Systems

Forward elimination ($i = 2, \dots, n$):

Define updated coefficients starting from $b_1^* = b$, $c_1^* = c$, $f_1^* = f_1$:

$$b_i^* = \frac{b}{a}b_{i-1}^* - c_{i-1}^*, \quad c_i^* = \frac{c}{a}b_{i-1}^*, \quad f_i^* = \frac{f_i}{a}b_{i-1}^* - f_{i-1}^*. \quad (24)$$

The reduced system is upper-bidiagonal:

$$b_i^*W_{i,j} + c_i^*W_{i+1,j} = f_i^*, \quad i = 1, \dots, n.$$

Back substitution:

$$W_{n,j} = \frac{f_n^*}{b_n^*}, \quad W_{i,j} = \frac{f_i^* - c_i^*W_{i+1,j}}{b_i^*}, \quad i = n-1, \dots, 1. \quad (25)$$

4.4 Solving the Recurrence for b_i^*

The recurrence in (24) is a second-order linear recurrence:

$$b_i^* - \frac{b}{a}b_{i-1}^* + \frac{c}{a}b_{i-2}^* = 0.$$

Its characteristic equation is:

$$\lambda^2 - \frac{b}{a}\lambda + \frac{c}{a} = 0 \implies \lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Define $L_i = \lambda_1^i - \lambda_2^i$. Then:

$$b_i^* = \frac{a^2}{\sqrt{b^2 - 4ac}} L_{i+1}.$$

4.5 Jump-Process Interpretation of the Implicit Scheme

Brennan & Schwartz (1978) show that as the grid becomes fine ($n, i \rightarrow \infty$), the inverse $\mathbf{A}^{-1}$ defines generalised “probabilities” p_ℓ^* such that:

$$W_{i,j} = (1 - rk) \sum_{\ell=-\infty}^{\infty} p_\ell^* W_{i+\ell, j+1}.$$

Theorem 4.1 (Brennan–Schwartz 1978). *Under condition $h^2 \leq \sigma^4/(r - \frac{1}{2}\sigma^2)^2$, all limiting probabilities $p_\ell^* \geq 0$, and*

$$\mathbb{E}[dy] = \left(r - \frac{1}{2}\sigma^2\right)k, \quad \text{Var}[dy] = \sigma^2k + \left(r - \frac{1}{2}\sigma^2\right)^2k^2. \quad (26)$$

The variance overestimates the GBM variance by a term of order k^2 , which vanishes as $k \rightarrow 0$.

4.6 Unconditional Stability of the Implicit Scheme

Implicit FDM: Unconditionally Stable

The implicit method is **unconditionally stable**: it produces bounded solutions for *any* choice of h and k . This is its main practical advantage over the explicit scheme.

5 Crank–Nicolson Method

5.1 Motivation: Second-Order Time Accuracy

Both explicit and implicit schemes are $O(k) + O(h^2)$. Crank–Nicolson achieves $O(k^2) + O(h^2)$ by averaging the spatial operators at levels j and $j + 1$.

5.2 Derivation

Satisfy the PDE at the *mid-point* in time $(i\Delta x, (j + \frac{1}{2})\Delta t)$ and approximate $[u_{xx}]^{j+1/2}$ by the arithmetic mean of its finite-difference approximations at levels j and $j + 1$:

$$\frac{U_i^{j+1} - U_i^j}{\Delta t} = \frac{1}{2} \left\{ \frac{U_{i+1}^{j+1} - 2U_i^{j+1} + U_{i-1}^{j+1}}{(\Delta x)^2} + \frac{U_{i+1}^j - 2U_i^j + U_{i-1}^j}{(\Delta x)^2} \right\}.$$

Setting $r_{CN} = \Delta t / (\Delta x)^2$ and rearranging:

$$-r_{CN}U_{i+1}^{j+1} + (2 + 2r_{CN})U_i^{j+1} - r_{CN}U_{i-1}^{j+1} = r_{CN}U_{i+1}^j + (2 - 2r_{CN})U_i^j + r_{CN}U_{i-1}^j. \quad (27)$$

5.3 The θ -Method (General Framework)

θ -Method

$$\frac{U^{j+1} - U^j}{\Delta t} = (1 - \theta) f(U^j) + \theta f(U^{j+1}), \quad 0 \leq \theta \leq 1. \quad (28)$$

θ	Method	Stability
0	Forward Euler (explicit)	Conditional
1/2	Crank–Nicolson	Unconditional
1	Backward Euler (implicit)	Unconditional

5.4 Accuracy Comparison via the Model Problem

Consider $u_t = -au$ ($a > 0$), exact solution $u(t) = u_0 e^{-at}$. The θ -method gives:

$$U^{j+1} = \lambda U^j, \quad \lambda = \frac{1 - (1 - \theta)ak}{1 + \theta ak}.$$

Compare with the exact amplification e^{-ak} via Taylor expansion (for small ak):

$$\lambda \approx 1 - ak + \theta(ak)^2 - \theta^2(ak)^3 + O((ak)^4),$$

$$e^{-ak} \approx 1 - ak + \frac{(ak)^2}{2} - \frac{(ak)^3}{6} + O((ak)^4).$$

Subtracting:

$$e^{-ak} - \lambda = \left(\frac{1}{2} - \theta\right)(ak)^2 + \left(\frac{1}{6} - \theta + \theta^2\right)(ak)^3 + O((ak)^4). \quad (29)$$

Crank–Nicolson Is Second-Order Accurate in Time

- $\theta \neq \frac{1}{2}$: leading error is $O((ak)^2) \Rightarrow$ *first-order* in time.
- $\theta = \frac{1}{2}$ (Crank–Nicolson): leading term vanishes, error is $O((ak)^3) \Rightarrow$ *second-order* in time.

6 Leapfrog Method and Its Instability

6.1 Leapfrog Discretisation

The leapfrog scheme uses a *centred* difference in *both* time and space:

$$\frac{U_i^{j+1} - U_i^{j-1}}{2\Delta t} = f(U^j). \quad (30)$$

Applied to $u_t = -au$:

$$U^{j+1} - U^{j-1} = -2ak U^j \quad (k = \Delta t).$$

6.2 Characteristic Equation

Substituting the Ansatz $U^j = \phi^j$:

$$\phi^2 + 2ak\phi - 1 = 0 \implies \phi_{1,2} = -ak \pm \sqrt{(ak)^2 + 1}.$$

General solution: $U^j = A\phi_1^j + B\phi_2^j$.

6.3 Proof of Unconditional Instability

Theorem 6.1. *When the leapfrog scheme (30) is applied to $u_t = -au$ with $a > 0$, it is **unconditionally unstable**.*

Proof. We have

$$\phi_2 = -ak - \sqrt{(ak)^2 + 1}.$$

For any $a > 0, k > 0$:

$$|\phi_2| = ak + \sqrt{(ak)^2 + 1} > 1.$$

For small ak :

$$\begin{aligned} \phi_1 &\approx 1 - ak \approx e^{-ak} \quad (\text{near true solution}), \\ \phi_2 &\approx -(1 + ak) \approx -e^{ak} \quad (|\phi_2| > 1, \text{ growing!}). \end{aligned}$$

Initial conditions $U^0 = u_0$, $U^1 \approx u_0 e^{-ak}$ produce a non-zero coefficient $B \neq 0$ for the ϕ_2 mode. Therefore $|U^j| \rightarrow \infty$ as $j \rightarrow \infty$, regardless of k . $\square$

Do NOT Use Leapfrog for Parabolic PDEs!

The leapfrog scheme is unconditionally unstable for parabolic equations (heat equation, Black–Scholes). It should never be used in option pricing.

7 Consistency, Stability, and Convergence

7.1 Three Fundamental Concepts

Definition 7.1 (Truncation Error & Consistency). Let $L(u) = 0$ be the PDE and $F(U) = 0$ the finite-difference scheme. The *truncation error* at node (ih, jk) is:

$$T_i^j(u) = F_i^j(u) - L_i^j(u).$$

The scheme is **consistent** if $T_i^j \rightarrow 0$ as $h, k \rightarrow 0$.

Definition 7.2 (Stability). The scheme is **stable** if the numerical solution does not grow unboundedly: there exists $M > 0$ (independent of h, k) such that $\|U^j\| \leq M\|U^0\|$ for all $j \geq 0$.

Definition 7.3 (Convergence). The scheme is **convergent** if $\max_{i,j} |U_i^j - u(ih, jk)| \rightarrow 0$ as $h, k \rightarrow 0$.

7.2 Lax–Wendroff Theorem

Theorem 7.1 (Lax–Wendroff Equivalence Theorem). *Suppose the continuous problem is well-posed (solution exists, is unique, and depends continuously on data), and the finite-difference scheme is consistent. Then:*

$$\text{Stability} \iff \text{Convergence}.$$

Equivalently: Consistency + Stability $\Rightarrow$ Convergence.

Practical Checklist

To prove that a scheme converges, it suffices to:

Step 1: Compute the truncation error via Taylor expansion and verify $T_i^j \rightarrow 0$ (*consistency*).

Step 2: Show $|G| \leq 1$ via von Neumann analysis (*stability*).

Step 3: Conclude convergence by Lax–Wendroff.

7.3 Consistency of Explicit, Implicit, Crank–Nicolson

For the heat equation $u_t = \alpha u_{xx}$, expand the explicit scheme at the exact solution u :

$$\frac{u(x, t+k) - u(x, t)}{k} = \alpha \frac{u(x+h, t+k) - 2u(x, t+k) + u(x-h, t+k)}{h^2}.$$

By Taylor expansion in k and h :

$$T = (u_t - \alpha u_{xx}) + \frac{k}{2} u_{tt} - \frac{\alpha h^2}{12} u_{xxxx} + O(k^2, h^4).$$

Since the exact solution satisfies the PDE, the bracketed term vanishes and $T = O(k) + O(h^2) \rightarrow 0$: *the explicit scheme is consistent.*

Similar analysis for Crank–Nicolson yields $T = O(k^2) + O(h^2)$.

8 von Neumann Fourier Stability Analysis

8.1 Principle

Assume the error decomposes into Fourier modes:

$$\varepsilon_i^j = G^j e^{i\xi x_i}, \quad i = \sqrt{-1}, \xi \in \mathbb{R}.$$

$G = G(\xi)$ is the *amplification factor*. The scheme is stable (in the ℓ^2 sense) if and only if:

$$|G(\xi)| \leq 1 \quad \forall \xi.$$

8.2 Explicit Scheme (Heat Equation)

For $u_t = \alpha u_{xx}$, the explicit scheme gives:

$$U_i^{j+1} = \lambda U_{i+1}^j + (1 - 2\lambda)U_i^j + \lambda U_{i-1}^j, \quad \lambda = \frac{\alpha k}{h^2}.$$

Substituting $\varepsilon_i^j = G^j e^{i\xi h}$:

$$G = \lambda e^{i\xi h} + (1 - 2\lambda) + \lambda e^{-i\xi h} = (1 - 2\lambda) + 2\lambda \cos(\xi h) = 1 - 4\lambda \sin^2\left(\frac{\xi h}{2}\right).$$

For $|G| \leq 1$:

$$-1 \leq 1 - 4\lambda \sin^2\left(\frac{\xi h}{2}\right) \leq 1.$$

The right inequality is trivially satisfied ($\lambda \geq 0$). The left gives $4\lambda \leq 2$, hence:

CFL Condition for Explicit Heat Equation

$$\lambda = \frac{\alpha k}{h^2} \leq \frac{1}{2} \iff k \leq \frac{h^2}{2\alpha}. \quad (31)$$

8.3 Explicit Scheme for the BS Equation

For scheme (15), define:

$$\alpha_0 = \frac{1}{2} \left(\frac{\sigma}{h} \right)^2 k, \quad \beta_0 = \frac{k}{2h} \left(r - \frac{1}{2} \sigma^2 \right),$$

so $a = \alpha_0 - \beta_0$, $b = 1 - 2\alpha_0$, $c = \alpha_0 + \beta_0$.

Substituting $\varepsilon_i^j = G^j e^{i\xi h}$:

$$\begin{aligned} G &= a e^{-i\xi h} + b + c e^{i\xi h} \\ &= 2\alpha_0 \cos(\xi h) + 2i\beta_0 \sin(\xi h) + (1 - 2\alpha_0). \end{aligned}$$

$$|G|^2 = (1 - 2\alpha_0 + 2\alpha_0 \cos \theta)^2 + 4\beta_0^2 \sin^2 \theta, \quad \theta = \xi h.$$

In the worst case ($\theta = \pi/2$, $\sin \theta = 1$):

$$|G|^2 = (1 - 2\alpha_0)^2 + 4\beta_0^2 \leq 1.$$

One can show this is equivalent to conditions (18).

8.4 Implicit Scheme (Heat Equation)

The implicit scheme gives:

$$U_i^j = (1 + 2\lambda)^{-1} (\lambda U_{i+1}^{j+1} + U_i^{j+1} + \lambda U_{i-1}^{j+1}) \implies G = \frac{1}{1 + 4\lambda \sin^2(\xi h/2)}.$$

Since the denominator ≥ 1 , we have $|G| \leq 1$ for all $\lambda > 0$: *unconditionally stable*.

8.5 Crank–Nicolson Scheme (Heat Equation)

From (27):

$$G = \frac{1 - 2r_{CN} \sin^2(\xi h/2)}{1 + 2r_{CN} \sin^2(\xi h/2)}. \quad (32)$$

Since both numerator and denominator have the form $1 \pm \delta$ ($\delta \geq 0$), we get $|G| \leq 1$ for all $r_{CN} > 0$: *Crank–Nicolson is unconditionally stable*.

8.6 Leapfrog Scheme: von Neumann Analysis

For the leapfrog PDE $u_t = f(u)$ applied to the heat equation, the amplification factor satisfies a *second-order* equation in G :

$$(2r + 1)G^2 - 4r \cos(\xi h) G + (2r - 1) = 0.$$

The product of roots equals $(2r - 1)/(2r + 1) < 1$ for $r > 0$, but the analysis of individual roots shows $|\Lambda_2| > 1$ when $r > \frac{1}{2}$ (i.e. $4r \sin^2(\xi h/2) > 2$). Even for $r < \frac{1}{2}$ the “parasitic” root $|\Lambda_2| = 1$ grows in time due to round-off, confirming unconditional instability for parabolic equations.

9 Binomial / Trinomial Trees and FDM

9.1 CRR Binomial Tree (Review)

In the Cox–Ross–Rubinstein (CRR) model with time step Δt :

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Convergence conditions (first- and second-order moment matching of $\ln S_{t+\Delta t} | \mathcal{F}_t$):

First-order moment. $p \ln u + (1 - p) \ln d = (r - \frac{1}{2}\sigma^2)\Delta t$ (to order Δt). □ □

Second-order moment. $p(\ln u)^2 + (1 - p)(\ln d)^2 - [p \ln u + (1 - p) \ln d]^2 = \sigma^2 \Delta t$ (to order Δt). □ □

9.2 Trinomial Tree

In a trinomial tree, at each step the log-price jumps by $+h$, 0 , or $-h$ with risk-neutral probabilities p^+, p, p^- :

$$p^- + p + p^+ = 1, \quad h(p^+ - p^-) = (r - \frac{1}{2}\sigma^2)k, \quad h^2(p^+ + p^-) = \sigma^2 k + O(k^2).$$

Comparing with the explicit FDM coefficients a, b, c from (16):

Exact Equivalence

$$p^- = a, \quad p = b, \quad p^+ = c.$$

The explicit FDM on log-price $y = \ln S$ is *exactly* a trinomial tree. The stability conditions (18) are precisely the non-negativity conditions for the jump

probabilities.

9.3 Convergence of Tree Models to GBM

Under $\Delta t \rightarrow 0$ with $n\Delta t = T$ fixed, we verify both moment conditions hold automatically for the explicit FDM coefficients.

Mean (first moment):

$$E[\Delta y] = h(c - a) = h \cdot \frac{k}{h} \left(r - \frac{1}{2}\sigma^2 \right) = \left(r - \frac{1}{2}\sigma^2 \right) k. \checkmark$$

Variance (second moment):

$$\text{Var}[\Delta y] \approx h^2(a + c) = h^2 \cdot \frac{\sigma^2 k}{h^2} = \sigma^2 k. \checkmark$$

By the central limit theorem, the trinomial tree converges in distribution to GBM as $\Delta t \rightarrow 0$.

10 Summary and Comparison

Scheme	Time acc.	Space acc.	Stability	Tree analogue	Cost/step
Explicit FDM	$O(k)$	$O(h^2)$	Conditional	Trinomial	$O(n)$
Implicit FDM	$O(k)$	$O(h^2)$	Unconditional	Generalised jump	$O(n)$
Crank–Nicolson	$O(k^2)$	$O(h^2)$	Unconditional	—	$O(n)$
Leapfrog	$O(k^2)$	$O(h^2)$	Unstable!	—	$O(n)$

Practical Recommendations

1. **Best accuracy & stability** $\rightarrow$ Crank–Nicolson.
2. **Simple implementation, small grid** $\rightarrow$ Explicit FDM (respect the CFL condition).
3. **Never use Leapfrog** for parabolic PDEs (BS equation, heat equation).
4. For American options: use the implicit or Crank–Nicolson scheme with the early-exercise constraint applied at each time step.

11 Homework Problems

Problem 1 – Finite Difference Approximations

- (a) Derive the central-difference approximation for $f''(x_0)$, equation (11), starting from Taylor expansions, and show explicitly that the truncation error is $O(h^2)$.
- (b) For $f(x) = e^x$ at $x_0 = 1$, numerically compute $f'(x_0)$ using forward, backward, and central differences with $h = 0.1, 0.01, 0.001$. Compare with the exact value e^1 and verify the convergence rates $O(h)$, $O(h)$, $O(h^2)$ respectively by computing the error ratios.

Problem 2 – Explicit FDM Stability

Given: $\sigma = 0.25$, $r = 0.05$.

- (a) Compute the maximum allowable h and k for the explicit finite difference scheme using conditions (18).
- (b) If $h = 0.1$ is fixed, what is the maximum time step k ?
- (c) Using von Neumann analysis, write down the amplification factor $G(\xi)$ for the explicit BS scheme. For which values of α_0 and β_0 does $|G| > 1$?
- (d) Implement the explicit FDM in Python/MATLAB to price a European call with $S_0 = 100$, $K = 100$, $T = 1$. Compare with the Black–Scholes formula. What happens when you violate condition (18)?

Problem 3 – Leapfrog Instability (Proof)

Apply the leapfrog scheme $\frac{U^{j+1} - U^{j-1}}{2k} = -aU^j$ to the model ODE $u_t = -au$ ($a > 0$).

- (a) Derive the characteristic equation and find both roots $\phi_{1,2}$.
- (b) Prove rigorously that $|\phi_2| > 1$ for all $ak > 0$.
- (c) Given initial conditions $U^0 = 1$, $U^1 = e^{-ak}$, find the coefficients A, B in $U^j = A\phi_1^j + B\phi_2^j$ and show $B \neq 0$. Conclude that $|U^j| \rightarrow \infty$ as $j \rightarrow \infty$.
- (d) Contrast with the true solution $u(t) \rightarrow 0$ as $t \rightarrow \infty$. What does this tell you about using leapfrog for option pricing?

Problem 4 – Crank–Nicolson Stability and Accuracy

- (a) Carry out the full von Neumann analysis for the Crank–Nicolson scheme (27) applied to $u_t = u_{xx}$. Derive the amplification factor G (equation (32)) and prove $|G| \leq 1$ for all $r_{CN} > 0$.
- (b) Using the θ -method (28) applied to $u_t = -au$, derive the error expression (29) in full. For which value of θ is the leading-order error minimised?
- (c) Show that the Crank–Nicolson scheme for the heat equation has truncation error $O(k^2) + O(h^2)$ by Taylor expansion.

Problem 5 – Tree-FDM Connection

(a) Verify that the CRR binomial tree satisfies both moment conditions (mean and variance of $\ln S_{t+\Delta t}$) to first order in Δt using the CRR parameters $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$.

(b) Show that the explicit FDM coefficients a, b, c from (16) satisfy the trinomial tree moment conditions:

$$h(c - a) = \left(r - \frac{1}{2}\sigma^2\right)k, \quad h^2(a + c) = \sigma^2k.$$

(c) Explain geometrically why the CFL condition $k \leq h^2/\sigma^2$ is equivalent to requiring $p^-, p, p^+ \geq 0$ in the trinomial tree.

(d) The JR (Jarrow–Rudd) model assumes $p = \frac{1}{2}$:

$$u = e^{(r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}}, \quad d = e^{(r - \frac{1}{2}\sigma^2)\Delta t - \sigma\sqrt{\Delta t}}.$$

Verify this also satisfies both moment conditions and discuss its relation to the explicit FDM.

Problem 6 – Numerical Implementation (Programming)

Implement and compare the following methods for pricing a European call with $S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.2$:

(a) Explicit FDM (ensure stability).

(b) Implicit FDM (Gaussian elimination, tridiagonal solver).

(c) Crank–Nicolson.

(d) CRR binomial tree.

For each method, plot the error versus the number of grid points n on a log–log scale. What convergence orders do you observe? Compare with the theoretical predictions from the truncation error analysis.

Hint: For parts (a)–(c), set $y_{\min} = \ln(S_0) - 5$, $y_{\max} = \ln(S_0) + 5$, use $n = 50, 100, 200, 400$ spatial points.

A Appendix A: Derivation of the Implicit Scheme Coefficients

Starting from (5) with the implicit discretisations (19)–(21):

$$\frac{1}{2}\sigma^2 \cdot \frac{W_{i+1,j} - 2W_{i,j} + W_{i-1,j}}{h^2} + \left(r - \frac{1}{2}\sigma^2\right) \cdot \frac{W_{i+1,j} - W_{i-1,j}}{2h} + \frac{W_{i,j+1} - W_{i,j}}{k} - rW_{i,j+1} = 0.$$

Multiply by k and collect terms:

$$\begin{aligned} \left(\frac{1}{k} - r\right) W_{i,j+1} &= \left[-\frac{\sigma^2}{2h^2} + \frac{r - \frac{1}{2}\sigma^2}{2h} + \frac{1}{k}\right] W_{i,j} \\ &+ \left[-\frac{\sigma^2}{2h^2} - \frac{r - \frac{1}{2}\sigma^2}{2h}\right] W_{i+1,j} \\ &+ \left[-\frac{\sigma^2}{2h^2} + \frac{r - \frac{1}{2}\sigma^2}{2h}\right] W_{i-1,j}. \end{aligned}$$

Multiplying through by k and rearranging into the form $aW_{i-1,j} + bW_{i,j} + cW_{i+1,j} = (1 - rk)W_{i,j+1}$ (note the sign convention in (22) uses $+rk$ with backward-in-time orientation) yields the coefficients (23).

B Appendix B: Full Proof of Stability for the θ -Method

For $u_t = -au$ the θ -method gives amplification $\lambda = (1 - (1 - \theta)ak)/(1 + \theta ak)$. Stability requires $|\lambda| \leq 1$.

Right bound $\lambda \leq 1$:

$$1 - (1 - \theta)ak \leq 1 + \theta ak \iff (1 - 2\theta)ak \leq 0.$$

For $a, k > 0$ this holds when $\theta \geq \frac{1}{2}$. For $\theta < \frac{1}{2}$ it holds whenever $ak \leq 2/(1 - 2\theta)$.

Left bound $\lambda \geq -1$:

$$1 - (1 - \theta)ak \geq -(1 + \theta ak) \iff 2 \geq (1 - 2\theta)ak.$$

Stability Summary for θ -Method

- $\theta \geq \frac{1}{2}$: unconditionally stable.
- $0 \leq \theta < \frac{1}{2}$: conditionally stable, $ak \leq \frac{2}{1 - 2\theta}$.

In particular, $\theta = 0$ (explicit) requires $ak \leq 2$; for the heat equation in space this translates to the CFL condition (31).

C Appendix C: Brennan–Schwartz (1978) – Key Results

Reference: M. J. Brennan & E. S. Schwartz, “Finite Difference Methods and Jump Processes Arising in the Pricing of Contingent Claims: A Synthesis,” *J. Financial & Quantitative Analysis*, 13(3), 461–474, 1978.

Main contributions:

1. Showed that solving the BS PDE by explicit FDM is *numerically equivalent* to approximating the GBM by a trinomial jump process (Cox & Ross 1976).
2. Showed that the implicit FDM corresponds to a more general jump process; the resulting “probabilities” are non-negative under mild conditions.

3. Established stability and accuracy conditions for both schemes (equations (18) and (26)).